

Product Requirements Document

Sentinel: Adaptive Hygiene Refund Verification Platform

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Role: Product Manager

Project Type: Trust & Safety Platform

Domain: Food Delivery

Duration: Concept Project

Platforms: Mobile Application + Internal Operations Dashboard

Executive Summary

Sentinel is a trust-based hygiene refund verification system designed to reduce fraudulent refund claims while preserving a seamless experience for legitimate customers.

The current refund process relies heavily on static image evidence and a frictionless verification workflow. While this approach optimizes customer experience, it creates vulnerabilities that can be exploited through recycled images, manipulated evidence, and emerging AI-assisted fraud techniques.

Sentinel introduces adaptive verification, where the level of verification required depends on user trust signals and behavioral risk indicators. Trusted users continue receiving a fast and frictionless experience, while higher-risk claims undergo enhanced verification through secure evidence collection and review workflows.

The objective is to reduce refund fraud, improve restaurant partner trust, and create a scalable trust and safety framework.

Problem

The current hygiene refund process lacks sufficient verification controls.

Users can upload images from their gallery and receive refunds with minimal validation.

This creates opportunities for:

- Recycled image abuse
- Manipulated evidence
- AI-generated contamination claims
- Organized refund exploitation

Business Impact:

- Increased refund leakage
- Declining restaurant partner trust
- Rising manual review costs
- Reduced confidence in automated refund systems

Evidence:

- Internal audits indicate 22% of hygiene refunds involve reused or recycled images.
- Restaurant Partner NPS declined from +40 to +12 YoY.
- 65% of partner complaints cite unjustified hygiene refunds.
- 12% of hygiene refund requests originate from serial refunders.

Why Now

The cost of creating fraudulent evidence is decreasing rapidly.

Consumer-grade AI tools and editing software have significantly lowered the barrier to generating convincing fake evidence.

At the same time:

- Refund abuse is increasing.
- Restaurant trust is decreasing.
- Fraud detection costs are rising.

Without intervention, fraud sophistication will continue growing faster than verification capabilities.

Goals

- Reduce fraudulent hygiene refund spend by 30% within 90 days.
- Reduce restaurant dispute volume by 25%.
- Maintain greater than 90% claim completion rate for trusted customers.
- Improve restaurant partner satisfaction.
- Reduce operational review burden.

Non-Goals

- Eliminating all refund fraud categories.
- Handling missing item or delivery-related complaints.
- Replacing human reviewers.
- Real-time AI analysis of submitted videos during Phase 1.

Target Users

High Trust Customers

Users with:

- Strong order history
- Low refund frequency
- High platform trust score

Medium Risk Customers

Users with:

- Limited trust history
- Moderate refund activity
- Inconsistent behavioral signals

High Risk Customers

Users with:

- High refund frequency
- Linked devices
- Suspicious behavioral patterns
- Previous abuse indicators

Trust Analysts

Internal operations teams responsible for fraud review and dispute resolution.

Restaurant Partners

Businesses impacted by refund decisions and chargebacks.

Assumptions

- Metadata serves as a supporting verification signal and should be evaluated alongside behavioral, device, and trust indicators.
- Risk scoring can be completed within 300 milliseconds.
- Existing infrastructure can support video verification storage requirements.
- Guided video verification increases the cost of fraud.
- Adaptive verification will not significantly impact trusted user satisfaction.

Solution Overview

Sentinel introduces a risk-based verification framework.

When a customer reports a hygiene issue, a Risk Engine evaluates trust and behavioral signals.

Based on the risk assessment, users are assigned to one of three verification experiences.

High Trust

Verification Experience:

- Gallery Upload
- Camera Upload

Goal:

Preserve speed and loyalty.

Outcome:

Instant refund workflow.

Medium Risk

Verification Experience:

- Secure Camera Only
- Metadata Capture

Goal:

Reduce manipulated evidence.

Outcome:

Enhanced automated review.

High Risk

Verification Experience:

- Guided Video Verification
- Metadata Validation
- Additional Review

Goal:

Prevent sophisticated fraud attempts.

Outcome:

Trust review workflow.

Product Principles

Principle 1: Verification Friction Scales with Risk

Verification requirements should adapt to user trust signals and behavioral risk indicators. Legitimate customers should experience minimal friction, while higher-risk claims undergo additional verification.

Principle 2: Protect Legitimate Customers First

Fraud prevention should not come at the cost of customer trust. The system should minimize false positives and preserve a seamless experience for genuine users.

Principle 3: Human Review Remains Available

Automated verification supports decision-making but does not replace human judgment. Users must always have access to an appeals and manual review process.

Principle 4: Restaurant Trust Is a First-Class Outcome

Verification decisions should balance customer satisfaction with restaurant protection. The platform must reduce unjustified chargebacks and improve partner confidence in refund decisions.

Principle 5: Fraud Prevention Should Be Invisible to Legitimate Users

Most customers should never notice the fraud prevention system. Verification mechanisms should operate seamlessly in the background whenever possible.

Principle 6: Transparency Builds Confidence

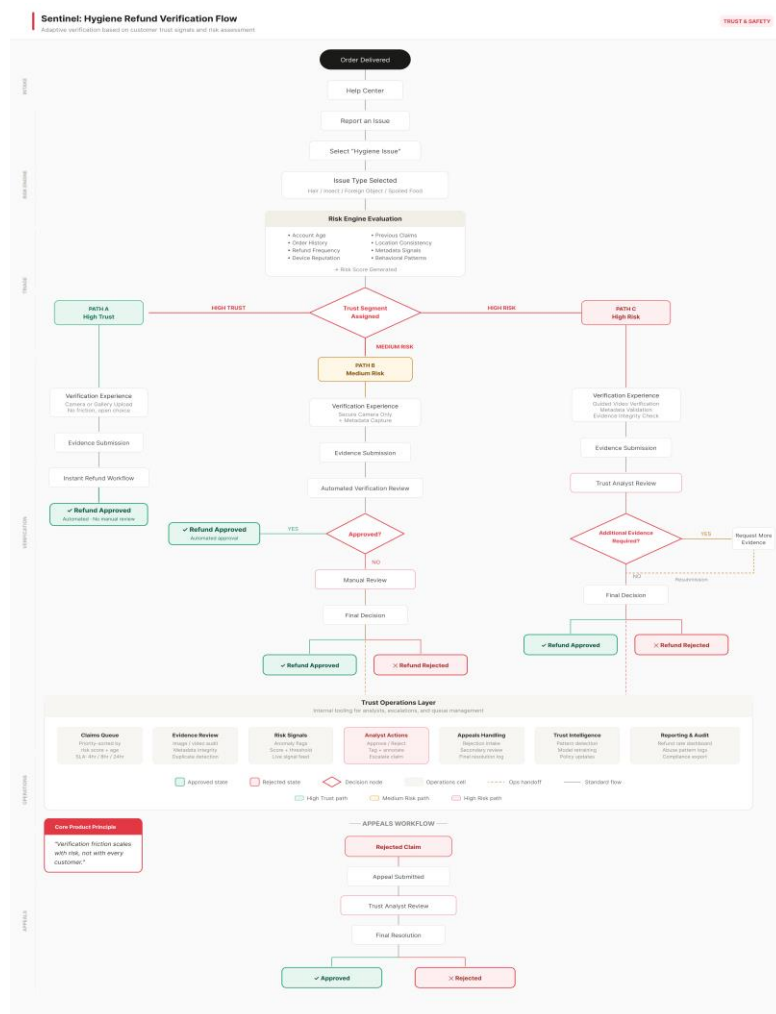
When additional verification is required, users should understand what is being requested and why. Clear communication reduces confusion, abandonment, and support dependency.

Principle 7: Operational Efficiency Matters

The system should not only prevent fraud but also help Trust & Safety teams investigate claims efficiently through prioritization, risk segmentation, and centralized review workflows.

USER FLOW DIAGRAM

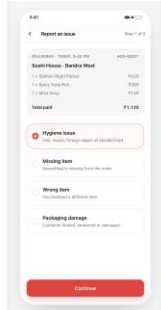
Sentinel uses adaptive verification to dynamically determine the appropriate evidence collection experience based on customer trust signals, behavioral indicators, and risk assessment. Most customers experience a frictionless journey, while higher-risk claims undergo enhanced verification and review workflows.



WIREFRAMES & EXPERIENCE DESIGN

Link: <https://claim-sentinel-guard.lovable.app/>

High Trust Experience:

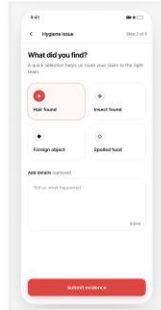


Screen 1:
Report an Issue.
Allow users to quickly select the issue category and initiate the refund process.

Key Components:

- Order Summary Card
- Delivered Items List
- Issue Selection Cards
- Primary CTA: Continue
- Step Indicator

Design Rationale:
The screen prioritizes issue identification with minimal cognitive effort. Issue cards use recognition rather than recall, enabling users to quickly select the most relevant problem type. The order summary provides contextual confidence before proceeding.

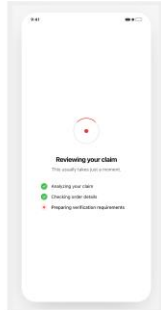


Screen 2:
Hygiene Issue Details.
Capture the specific hygiene issue and any supporting context.

Key Components:

- Hygiene Issue Selection Grid
- Optional Notes Field
- Character Counter
- Primary CTA: Submit Evidence

Design Rationale:
The predefined issue categories reduce ambiguity and improve downstream routing accuracy. Optional notes provide flexibility without forcing additional effort, maintaining a low-friction experience.

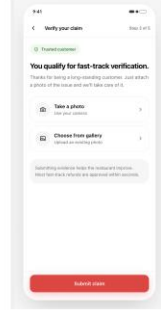


Screen 3:
Risk Evaluation.
Communicate that the system is reviewing claim information before determining verification requirements.

Key Components:

- Loading Indicator
- Verification Progress Messages
- Status Checklist

Design Rationale:
This screen creates transparency around the verification process while masking backend latency. Progressive status updates reassure users that the system is actively processing their claim.



Screen 4:
Fast-Track Verification.
Allow trusted customers to submit evidence using the quickest possible path.

Key Components:

- Trusted Customer Badge
- Camera Upload Option
- Gallery Upload Option
- Submit Claim CTA

Design Rationale:
High-trust customers are rewarded with a frictionless experience. Maintaining convenience for reliable users protects customer satisfaction while reinforcing positive platform behavior.

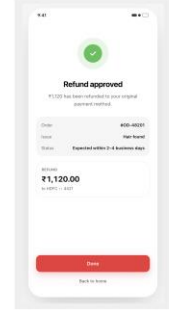


Screen 5:
Review Evidence.
Allow users to validate the selected evidence before submission.

Key Components:

- Evidence Preview
- Verification Status Card
- Retake Option
- Submit Claim CTA

Design Rationale:
Review screens reduce accidental submissions and increase user confidence. Visible verification indicators provide reassurance that evidence has been captured successfully.



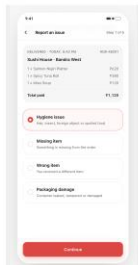
Screen 6:
Refund Approved.
Confirm successful claim resolution and communicate refund details.

Key Components:

- Success Indicator
- Refund Amount
- Refund Timeline
- Order Summary
- Done CTA

Design Rationale:
The final screen creates a positive emotional endpoint. Clear refund details reduce support inquiries while reinforcing trust in the platform.

Enhanced Verification Experience:

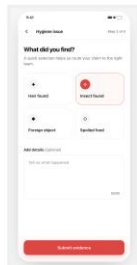


Screen 1:
Report an Issue.
Allow users to begin the hygiene complaint journey.

Key Components:

- Order Summary
- Issue Selection Cards
- Continue CTA

Design Rationale:
Consistent entry point ensures all users start from the same workflow regardless of risk profile.

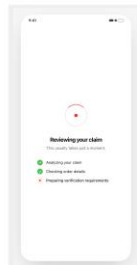


Screen 2:
Hygiene Issue Details.
Capture issue categorization and supporting information.

Key Components:

- Hygiene Issue Categories
- Notes Field
- Submit Evidence CTA

Design Rationale:
Standardized issue classification improves fraud detection accuracy and operational review efficiency.



Screen 3:
Risk Evaluation.
Determine the appropriate verification pathway based on risk signals.

Key Components:

- Loading Indicator
- Verification Progress Messages

Design Rationale:
Creates a seamless transition between issue reporting and verification while communicating system intelligence.



Screen 4:
Secure Camera Verification.
Require real-time evidence capture for higher-risk claims.

Key Components:

- Secure Camera Module
- Camera CTA
- Verification Information
- Security Reassurance Copy

Design Rationale:
Disabling gallery uploads reduces image reuse and manipulation risks. Real-time capture increases evidence authenticity while maintaining a simple user experience.



Screen 5:
Guided Video Verification.
Capture stronger evidence through a short guided video verification process.

Key Components:

- Full-Screen Camera Preview
- Recording Status
- Progress Indicator
- Motion Guidance Overlay
- Remaining Time Indicator

Design Rationale:
Guided movement increases evidence reliability and makes image manipulation significantly more difficult. The interaction is intentionally lightweight to minimize abandonment while increasing verification confidence.



Screen 6:
Review Evidence.
Allow users to verify captured video evidence before submission.

Key Components:

- Video Preview
- Verification Status Card
- Retake Option
- Submit Claim CTA

Design Rationale:
Providing a review step reduces accidental submissions and ensures users are comfortable with the evidence being submitted.



Screen 7:
Evidence Received.
Acknowledge successful claim submission and set review expectations.

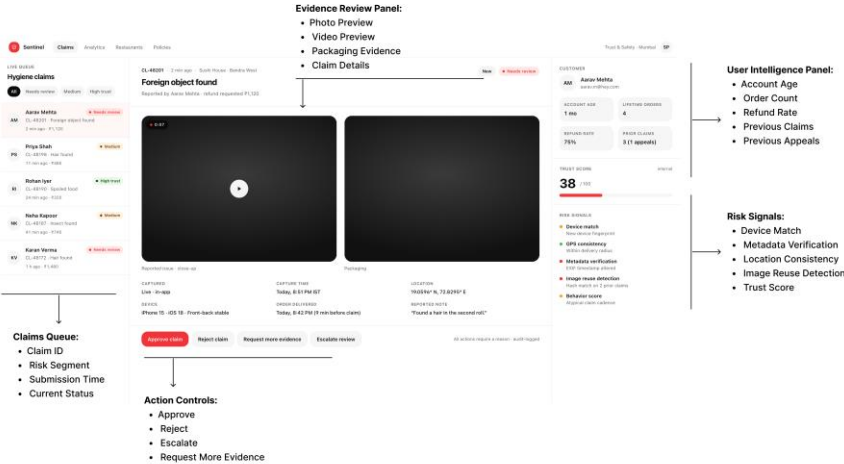
Key Components:

- Submission Confirmation
- Expected Review Timeline
- Claim Summary
- Done CTA

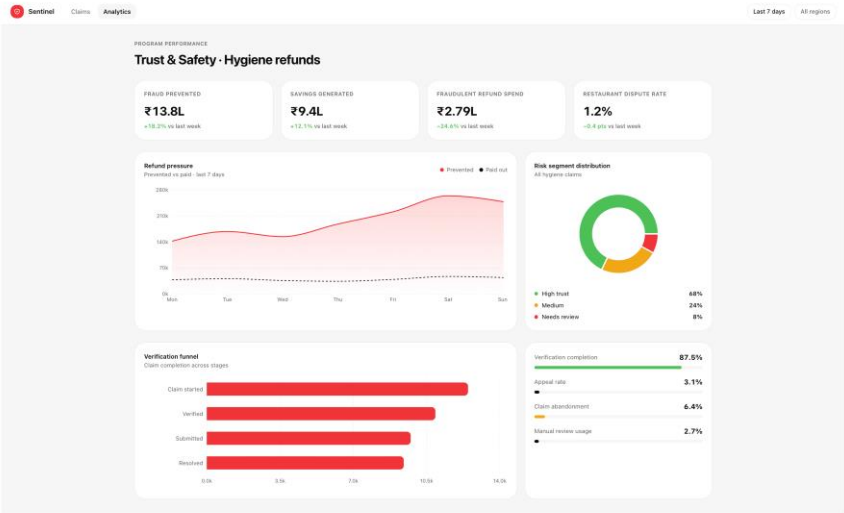
Design Rationale:
Higher-risk claims require additional review. Setting clear expectations reduces uncertainty and prevents unnecessary support contacts.

Trust Analyst Dashboard:

Objective:
Enable Trust & Safety teams to efficiently review, investigate, and resolve hygiene-related refund claims.



Design Rationale:
The dashboard provides analysts with all decision-making inputs within a single workspace. Centralizing evidence, risk signals, and user history reduces investigation time while improving decision quality and consistency.



User Stories

High Trust Customer

As a loyal customer,

I want hygiene issues resolved quickly, so that I continue trusting the platform.

Medium Risk Customer

As a customer reporting a genuine issue,

I want a simple verification experience, so that I can submit evidence easily.

High Risk Customer

As a customer requiring additional verification,

I want clear guidance throughout the process,
so that I can complete my claim successfully.

Restaurant Partner

As a restaurant partner,
I want fraudulent claims identified accurately,
so that I am not unfairly penalized.

Trust Analyst

As a trust analyst,
I want claims prioritized by risk,
so that investigations can be performed efficiently.

Functional Requirements

Risk Segmentation

- FR-1: System shall evaluate user risk when Hygiene Issue is selected.
- FR-2: System shall classify users into High Trust, Medium Risk, or High Risk.
- FR-3: Risk scoring shall be completed within 300ms.

Verification Experience

- FR-4: High Trust users shall continue through the standard refund workflow.
- FR-5: Medium Risk users shall use Secure Camera for verification.
- FR-6: Gallery uploads shall be disabled for Medium Risk users.
- FR-7: High Risk users shall complete Guided Video Verification.
- FR-8: Video duration shall be between 5–7 seconds.
- FR-9: Visual instructions shall be provided during recording.

Metadata Collection

- FR-10: Timestamp metadata shall be captured.

FR-11: Device metadata shall be captured.

FR-12: Location metadata shall be captured when permission is granted.

FR-13: Metadata shall be stored alongside submitted evidence.

Appeals & Reviews

FR-14: Users shall have access to manual review.

FR-15: Trust analysts shall be able to override automated outcomes.

FR-16: Approved appeals shall trigger trust profile recalculation.

Success Metrics

North Star Metric

Fraudulent Hygiene Refund Spend

Business Metrics

- Restaurant Partner NPS
- Chargeback Volume
- Restaurant Dispute Rate
- Refund Losses Prevented

Product Metrics

- Video Completion Rate
- Secure Camera Completion Rate
- Risk Score Accuracy
- Appeal Resolution Time
- Claim Completion Rate

Guardrail Metrics

- CSAT
- Claim Abandonment Rate
- Support Ticket Volume
- Refund Resolution Time

Analytics Instrumentation

User Journey Events

- claim_started
- issue_type_selected
- risk_api_called
- risk_segment_assigned
- secure_camera_opened
- photo_captured
- video_verification_started
- video_verification_completed
- claim_submitted
- refund_approved
- refund_rejected

Fraud Events

- recycled_image_detected
- metadata_missing
- location_mismatch_detected
- fraud_confirmed
- fraud_prevented

Partner Events

- restaurant_dispute_created
- restaurant_dispute_resolved

Experiment Plan

Experiment 1

Secure Camera reduces image reuse fraud.

Success Metric:

20% reduction in recycled-image claims.

Experiment 2

Guided Video Verification reduces fraudulent approvals.

Success Metric:

30% reduction in suspicious payouts.

Experiment 3

Adaptive verification preserves customer experience.

Success Metric:

No significant decline in High Trust customer CSAT.

Rollout Plan

Phase 0

Internal Validation

Phase 1

Shadow Mode

Phase 2

High Risk Rollout

Phase 3

Medium + High Risk Rollout

Phase 4

Full Rollout

Dependencies

Trust & Safety

Data Science

Mobile Engineering

Backend Engineering

Analytics

Legal & Compliance

Customer Support

Partner Operations

Risks & Trade-offs

- False positives
- User friction
- Storage costs
- Device limitations
- Model drift

Mitigations should be documented for each risk.

Launch Readiness Checklist

- ☐ Risk API validated
- ☐ Secure Camera tested
- ☐ Video upload success rate $\geq 95\%$
- ☐ Analytics verified
- ☐ Manual review operational
- ☐ Legal approval completed
- ☐ Monitoring dashboards configured

Open Questions

- What refund threshold should trigger High Risk classification?
- Should restaurant trust score influence verification requirements?
- What is the optimal evidence retention period?

Future Scope

Phase 2

- Device reputation graph
- AI artifact detection
- Cross-account fraud clustering

Phase 3

- Restaurant trust scoring
- Real-time fraud confidence scoring
- Multimodal verification

Phase 4

- Behavioral biometrics
- Fraud intelligence network
- Adaptive verification policies

Platform Vision

Sentinel is designed as a trust infrastructure layer rather than a standalone refund verification feature.

While the initial use case focuses on hygiene refunds; the underlying trust engine can evolve to support:

- Restaurant disputes
- Loyalty programs
- Promotions and incentives
- Account risk management
- Partner trust scoring

Long-term, Sentinel can become a shared trust platform that powers decision-making across the marketplace ecosystem.

Recommendation

Sentinel should be positioned as a trust platform rather than a fraud-prevention feature.

The objective is not to eliminate every fraudulent claim.

The objective is to make fraud increasingly difficult, expensive, and unreliable while preserving a fast and trustworthy experience for legitimate customers and restaurant partners.